First we need to import

- import pandas as pd
  import numpy as np
  import seaborn as sns
  import matplotlib.pyplot as plt

# ~~s~~. upload and read dataset,
df = ~~read~~ pd.read_csv(' winequality-red.cs

~~look~~ Now look and know about dataset
~~imp~~ information
df.head # to ~~a~~. know the heading
df.info to see information about dat
set
df.describe() ~~to~~ to see statistical
summary of the dataset

Date

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```
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import numpy as np
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import matplotlib.pyplot as plt
```

upload and read dataset,

df = ~~read~~ pd.read_csv(' winequality-red.csv)

~~look~~ Now look and know about dataset ~~imp~~ information

df.head # to know the heading

df.info to see information about data set

df.describe() to see statistical summary of the dataset

df.shape to see number of column in the dataset

df.tail to see last 5 row of the data set

step to move to second feature of EDA

Data cleanning

df.column() To see name in the column in the dataset.

df.isnull().sum() to check for missing value in the dataset

Date

check unique value

df['quality'].unique()
df['quality'].value_count()

step 3

understand the target variable
what are we trying to predict

target = quality

Next step - univariate Analysis

How are the values distributed
example
alchol
pH
sulphate
citric acid
volatile acidity

we start with alchol because it strongly affects wine quality

what we know about alchol
minimum alchol
maximum alchol
average alchol
distribution

df.['alcohol'].describe() # to see statistical summary of the acholo column

Next step ~~we~~

we ~~visuli~~ visualize alcohol distribution using Histograph

```
ptl(figuresize (10, 6))
sns. hisplot (df ['alcohol'], bins=30, kde = True)
plt . title ('Distribution of Alchol Content Redwine')
plt. xlabel ('Alcohol content')
plt . y label (' frequency)
```

observation

most of wines have alchol between 9.5 - 11.5

observation 2

High alchol wine 13 - 15 are rare

observation 3

most value are on the left
Few extreme value on the right

Step 5

Bivariate Analysis
feature vs Target
example :
Alcohol vs Quality

Question we want to answer
Do higer alcohol is better quality

for this we need to use box plot

```
plt. figure(figuresize = (20, 10))
sns. box plot( x = 'quality of alcohol', y = alcohol, data = df)
plt. title('Boxplot of Alcohol content by wine quality')
plt. xlabel('wine quality')
plt. ylabel('Alcohol content')
plt. show()
```

7 correlation Analysis
How the feature are storngly releated to each other and how storngly they relate to the target variable

Question like
Does alcohol increase quality of
Does volatile acidity reduce quality

we use a correlation heatmap

```
plt.figure(figuresize = 10, 8)
corr = df.corr()
sns.heatmap(corr, annot = True, cmap = coolwarm)
plt.title('corelation Heatmap of wine quality Dataset')
plt.show()
```

(8) outlier Detection

Now we check if there are extreme value

outlier can:

distort models

reduce prediction accuracy

most alchol values = 9-11

But some = 15

```
sns.boxplot(y = df['alochol'])
```

Feature engineering

Now we create new feature from existing data

Because better feature → better ML models

quality classification

instead classification

low quality = 3, 4, 5

High quality = 6, 7, 8